CSC2107: Design and Analysis of Algorithms

Group Assignment One due September 4, 2026

This covers the key concepts around data structures under arrays and dictionaries.

a) **Write programs to implement the following operations on arrays Using language of your choice C or Java or Python.**
   1. What are axiom to implement array operations?
   2. Traversal of single dimension, 2-dimention and 3-dimension arrays and determine the base address of each array.
   3. Create two single dimension arrays and merge them as a single array.
   4. Create a two-dimensional array, design two separate functions that return the minimum element and maximum element with their locations.

b) **Assignment: Air Pollution and Weather Data Analysis Using Arrays**

The AirQo Sensor Project at Makerere University is monitoring air pollution and weather conditions around Mukwano Industries in Kampala. During the monitoring period, sensor measurements were collected continuously and aggregated into monthly values. The table below presents the monthly PM2.5 concentration, temperature, and relative humidity recorded for one year.

Month	PM2.5 (μg/m³)	Temperature (°F)	Humidity (%)
January	23	40	26
February	31	43	25
March	34	48	29
April	42	53	38
May	40	64	42
June	42	68	38
July	27	75	34
August	23	48	42
September	27	48	38
October	26	60	32
November	23	53	31
December	31	40	26

Using **arrays**, store and process the monthly AirQo sensor data shown above. Write a program that:

1. Finds and displays the **month with the highest PM2.5 concentration**.
2. Calculates and displays the **average PM2.5 concentration** for the entire year.
3. Finds and displays the **month with the highest humidity**.
4. Calculates and displays the **average temperature** for the entire year.
5. Finds the **month with the lowest PM2.5 concentration**.
6. Displays all months where the **PM2.5 concentration was above the annual average**.
7. Using a Dictionary, store each month as a key and its corresponding PM2.5 concentration (μg/m³) as the value. Write a program that:
   I. Stores all the monthly PM2.5 measurements in the Dictionary.
   II. Prompts the user to enter a month and displays the corresponding PM2.5 concentration.
   III. Finds and displays the month with the highest PM2.5 concentration.
   IV. Calculates and displays the average PM2.5 concentration for all months.
8. Using Sets, store the monthly values for PM2.5, temperature, and humidity in three separate sets. Write a program that:
   I. Displays the unique values recorded for each of the three environmental variables.
   II. Determines the number of unique values recorded for PM2.5, temperature, and humidity.
   III. Identifies which of the three variables has the highest number of repeated values during the year.